

Total No. of Questions : 4]

SEAT No. :

PF-306

[Total No. of Pages : 2

Apr-26/TE/Insem-386

T.E. (Artificial Intelligence and Data Science) (Insem.)

ARTIFICIAL NEURAL NETWORK
(2019 Pattern) (Semester - VI) (317531)

Time : 1 Hour]

[Max. Marks : 30

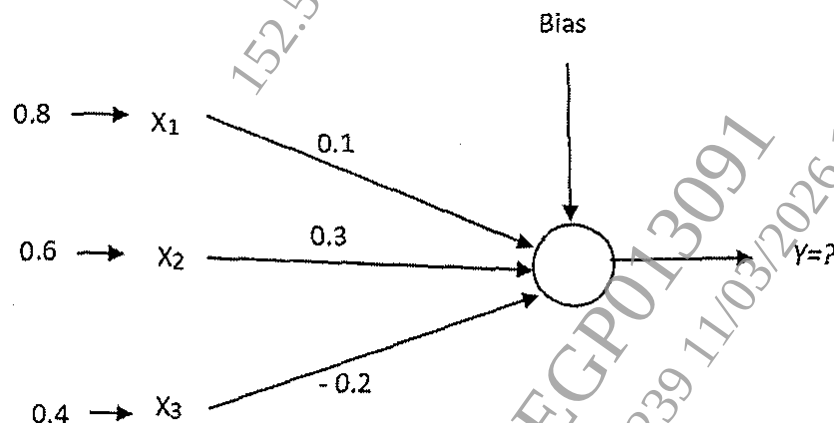
Instructions to the candidates :

- 1) Solve questions Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is neural network topology? Explain any Two topologies in detail. [5]
- b) Explain briefly Mc Culloch Pitt's (MP) artificial neuron model. Give its limitations. [5]
- c) Distinguish between Biological neural network and artificial neural networks. [5]

OR

- Q2)** a) Explain the structure and function of a biological neuron. [5]
- b) Obtain the output of the neuron Y for the network shown in figure using activation function as Binary sigmoidal. [5]



- c) What is a neural network activation function? Explain any one activation function in detail. [5]

P.T.O.

- Q3)** a) Write and explain perceptron learning algorithm. [5]
b) Draw and explain the architecture of multilayer feed forward networks. [5]
c) Differentiate between Learning and Memory. [5]

OR

- Q4)** a) Write and explain Hebbian learning Algorithm. [5]
b) What is error correction learning? Explain in detail with diagram. [5]
c) Differentiate between Feed Forward and Feedback neural network. [5]

